SAR-Based flood Mapping Using Machine Learning:
A Case Study in Porto Alegre, Brazil

Author: Carolina da silva Ezequiel
Github: https://github.com/carolezeq-Analist
Portfolio: http://carolina-geo-folio.lovable.app/

In May 2024, the state of Rio Grande do Sul, southern Brazil, experienced the most severe flood disaster in its recorded history. Driven by an extreme precipitation event associated with an extratropical cyclone, accumulated rainfall exceeded 400 mm in less than 72 hours across multiple river basins, triggering widespread inundation across the metropolitan region of Porto Alegre and surrounding municipalities. The event resulted in over 150 fatalities, the displacement of more than 400,000 people, and extensive damage to urban infrastructure, agricultural land, and transportation networks, consolidating itself as a landmark climate disaster in Brazil and reinforcing the urgency of robust geospatial monitoring frameworks for disaster response and risk management.

Rapid and accurate mapping of flood extent is a critical component of emergency response, damage assessment, and post-disaster recovery planning. Optical remote sensing satellites, such as Sentinel-2 and Landsat, have been widely used for land cover monitoring and vegetation analysis due to their high spatial resolution and multispectral capabilities. However, optical sensors are fundamentally constrained by atmospheric conditions: cloud cover, which is inherently persistent during intense precipitation events, prevents the acquisition of usable imagery precisely when flood monitoring is most needed. This limitation significantly reduces the operational utility of optical data during active flood scenarios, creating a critical observational gap at the moment of greatest demand.

Synthetic Aperture Radar (SAR) systems offer a technically robust solution to this constraint. Unlike passive optical sensors, SAR instruments operate in the microwave portion of the electromagnetic spectrum, emitting and receiving their own radar pulses independently of solar illumination or atmospheric cloud cover. This all-weather, day-and-night acquisition capability makes SAR particularly well-suited for emergency mapping applications. Water surfaces, characterized by specular reflection of radar signals, produce consistently low backscatter values in SAR imagery, generating a strong and reliable spectral contrast with surrounding land cover types such as vegetation and dry soil. The Sentinel-1 satellite mission, operated by the European Space Agency under the Copernicus Programme, provides C-band SAR data at 10-meter spatial resolution with a five-day revisit frequency, offering a freely available and operationally relevant data source for flood monitoring at regional scales.

Despite the recognized potential of SAR for flood detection, threshold-based classification approaches which define inundated areas based on fixed backscatter cutoff values remain sensitive to local environmental conditions, sensor geometry, and land cover heterogeneity, limiting their transferability across different flood events and geographic contexts. Machine learning classification methods, particularly Random Forest, have demonstrated superior performance in complex land cover discrimination tasks by learning non-linear relationships between input spectral features and target classes, offering a more adaptive and robust alternative to fixed-threshold approaches.

This study presents a SAR-based flood mapping methodology applied to the May 2024 Rio Grande do Sul flood event, integrating Sentinel-1 C-band imagery

processed in the Google Earth Engine cloud computing platform with a Random Forest supervised classifier. The proposed approach classifies the flood-affected landscape into three land cover classes: open water, vegetation, and dry soil and evaluates classification performance through accuracy assessment metrics including overall accuracy and Cohen's Kappa coefficient. The results demonstrate the operational viability of combining freely available SAR data with machine learning in a cloud-based environment for rapid flood extent mapping, with direct relevance to disaster risk reduction and climate adaptation strategies in Brazil and beyond.

The study area encompasses the metropolitan region of Porto Alegre, capital of the state of Rio Grande do Sul, located in southern Brazil (approximately 30°01'S, 51°13'W). The region is situated within the Guaíba Lake hydrographic basin, which receives contributions from five major river systems the Jacuí, Taquari, Caí, Sinos, and Gravataí rivers, draining a combined catchment area of approximately 200,000 km². The Guaíba Lake itself, despite its local designation as a lake, functions hydrologically as a wide fluvial channel connecting the upstream river network to the Lagoa dos Patos coastal lagoon system.

The metropolitan region is characterized by a subtropical humid climate (Köppen: Cfa), with no defined dry season and mean annual precipitation ranging between 1,200 and 1,500 mm, distributed relatively uniformly throughout the year. The landscape presents a marked spatial heterogeneity, comprising densely urbanized areas along the Guaíba shoreline and river margins, extensive agricultural and pastoral land in the surrounding municipalities, riparian forest fragments, and low-lying floodplain zones historically susceptible to periodic inundation.

During the May 2024 extreme precipitation event, the Guaíba Lake water level reached 5.33 meters on May 5th, 2024 surpassing the previous historical record of 4.76 meters registered during the catastrophic 1941 floods and remained above flood alert thresholds for over three weeks. The municipalities of Porto Alegre, Canoas, São Leopoldo, Guaíba, Eldorado do Sul, and Nova Santa Rita were among the most severely affected, with large proportions of their urban and peri-urban territories remaining submerged for extended periods.

Flood extent mapping was performed using Sentinel-1 Ground Range Detected (GRD) imagery acquired by the European Space Agency (ESA) under the Copernicus Programme. Sentinel-1 operates in C-band (5.405 GHz, wavelength approximately 5.6 cm) using an Interferometric Wide (IW) swath mode, providing imagery at 10-meter spatial resolution across a 250-km swath width. The constellation's five-day revisit frequency over South America ensures timely data availability for operational monitoring applications.

Images were filtered to include only descending orbit passes acquired between May 1st and May 12th, 2024, capturing the peak inundation period of the flood event. Both VV (vertical transmit, vertical receive) and VH (vertical transmit, horizontal receive) polarization channels were retained for analysis. The VV polarization is primarily sensitive to surface roughness and moisture conditions, while the VH cross-polarization channel provides complementary information on volumetric

scattering from vegetation canopy structures. The harmonized surface backscatter product (COPERNICUS/S1_GRD) was accessed directly through the Google Earth Engine cloud computing platform, eliminating the need for local data download and preprocessing.

Municipal boundary vectors for Porto Alegre were obtained from the Brazilian Institute of Geography and Statistics (IBGE) territorial mesh database and used to define the spatial extent of the analysis and to provide geographic context for cartographic outputs. All spatial analyses were performed in the SIRGAS 2000 / UTM Zone 22S coordinate reference system, consistent with Brazilian national cartographic standards.

Overview

The flood mapping workflow was implemented entirely within the Google Earth Engine (GEE) cloud computing platform, combining Sentinel-1 SAR preprocessing, vegetation index derivation, supervised machine learning classification, and accuracy assessment in a single integrated processing environment. The methodology consists of five sequential stages: (1) image collection and filtering, (2) backscatter conversion and index calculation, (3) training sample collection, (4) Random Forest classification, and (5) accuracy assessment and cartographic output generation. The complete processing script is publicly available at the corresponding author's GitHub repository.

Sentinel-1 GRD imagery accessed through GEE is provided in decibel (dB) scale, representing the logarithmic transformation of the radar backscatter coefficient. Although the dB representation is conventional in SAR analysis and facilitates visual interpretation, mathematical operations between bands required for vegetation index computation must be performed in linear power scale to avoid physically inconsistent results. Accordingly, all imagery was converted from dB to linear scale prior to index calculation using the standard transformation:

$$\sigma_linear = 10^{(\sigma_dB / 10)}$$

where σ_dB represents the backscatter value in decibel scale and σ_linear the corresponding linear power value. Following index computation, results were interpreted in their native linear scale without back-conversion to dB.

A modified Radar Vegetation Index (RVImod) was computed from the linear-scaled VV and VH polarization bands to enhance the spectral separability between land cover classes. The RVImod formulation used in this study follows:

$$RVImod = (4 \times VH) / (VV + VH)^2$$

This index exploits the differential sensitivity of VV and VH polarizations to surface scattering mechanisms: open water surfaces produce predominantly specular reflection, yielding very low backscatter in both channels but with distinct VV/VH ratios compared to vegetated or bare soil surfaces. The RVImod index amplifies these contrasts, providing an additional discriminating feature beyond the individual polarization channels and improving classifier performance in heterogeneous

landscapes. The final input feature stack for classification comprised three bands: VV, VH, and RVImod.

Reference training samples were collected through visual interpretation of the Sentinel-1 SAR composite displayed in the GEE mapping interface, supported by the satellite basemap for geographic context. Three land cover classes were defined for classification: water (class 0), vegetation (class 1), and dry soil (class 2). These classes were selected to reflect the dominant surface types present in the study area during the flood event and to maximize inter-class spectral separability in the SAR feature space.

Training points were collected as individual georeferenced point features distributed across the full spatial extent of the study area, avoiding transitional zones and mixed pixels at class boundaries to maximize training sample purity. A total of 206 training points were collected across the three classes: 68 water points, 71 vegetation points, and 67 dry soil points, providing a balanced class distribution that prevents classifier bias toward any dominant land cover type. Each point was assigned a numeric class label and merged into a unified FeatureCollection prior to pixel value extraction.

Pixel values for the three input features (VV, VH, RVImod) were extracted at each training point location using the sampleRegions function in GEE at 10-meter spatial resolution, with a tileScale parameter of 4 applied to manage computational memory requirements for large-area processing.

Random Forest Classification

The labeled sample dataset was randomly partitioned into training (80%) and validation (20%) subsets using a uniformly distributed random variable appended to each sample feature. This stratified random split ensures that both subsets are representative of the full range of spectral variability present in the training data, and that the validation subset remains independent from the training process.

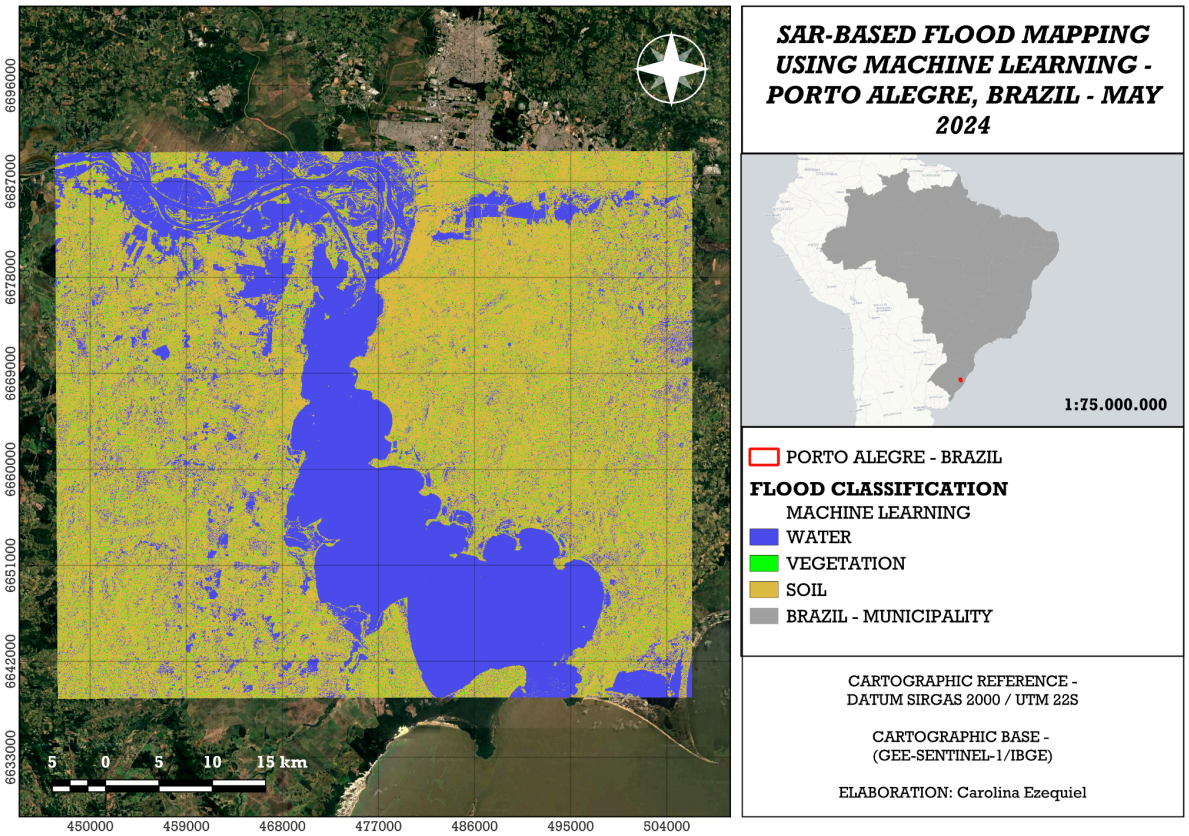
A Random Forest classifier was trained using the smileRandomForest implementation available in the GEE machine learning library, configured with 50 decision trees. Random Forest is an ensemble learning method that constructs multiple decorrelated decision trees during training, each built on a random bootstrap subsample of the training data and a random subset of input features at each node split. Final class predictions are determined by majority voting across all trees, reducing variance and improving generalization relative to single decision tree classifiers. The trained classifier was applied to the full three-band input image clipped to the study area boundary, producing a pixel-level land cover classification map at 10-meter spatial resolution.

Classification accuracy was evaluated using the independent 20% validation subset withheld during the training phase. Validation samples were classified using the trained Random Forest model and compared against their reference class labels to construct an error matrix. Two summary statistics were derived from the error matrix to quantify classification performance:

Overall Accuracy (OA) the proportion of correctly classified validation samples across all classes, expressed as a value between 0 and 1.
Cohen's Kappa Coefficient (κ), a statistical measure of agreement between predicted and reference classifications that accounts for the probability of chance agreement, providing a more conservative and robust performance estimate than overall accuracy alone. Kappa values above 0.80 are generally interpreted as strong agreement in remote sensing classification literature.

Cartographic Output

The final classified raster was exported from GEE to Google Drive in GeoTIFF format at 10-meter spatial resolution, georeferenced to the SIRGAS 2000 / UTM Zone 22S coordinate reference system. Cartographic layout production was performed in QGIS, incorporating the classification result overlaid on a Sentinel-1 SAR composite background. All cartographic elements follow standard mapping conventions, including north arrow, scale bar, coordinate grid, legend, and data source attribution. The final map was exported in high-resolution PNG format.



Classification Performance

The Random Forest classifier trained on 206 georeferenced sample points achieved an Overall Accuracy (OA) of 0.857 and a Cohen's Kappa coefficient (κ) of 0.742 on the independent 20% validation subset. These results indicate strong

classification performance, with the model correctly identifying land cover class in approximately 86% of validation samples and demonstrating substantial agreement beyond chance expectation. The Kappa value of 0.742 falls within the range conventionally interpreted as strong agreement in remote sensing classification literature, confirming the reliability of the trained model for operational flood mapping applications.

The error matrix derived from validation sample classification is presented in Table 1. Inspection of the matrix reveals that the water class (class 0) achieved perfect or near-perfect classification, with no misclassification to other land cover types, reflecting the strong and consistent low-backscatter signal produced by open water surfaces in C-band SAR imagery. The dry soil class (class 2) also demonstrated high classification accuracy, with minimal confusion with adjacent classes. The vegetation class (class 1) presented the highest inter-class confusion, with a subset of validation samples misclassified as dry soil. This pattern is consistent with known limitations of C-band SAR in discriminating sparse or low-biomass vegetation from bare soil surfaces, where the volumetric scattering contribution of the canopy may be insufficient to produce a clearly distinguishable backscatter signature relative to underlying mineral soil.

Table 1. Error matrix for Random Forest classification validation. Rows represent reference classes; columns represent predicted classes. Class labels: 0 = Water, 1 = Vegetation, 2 = Dry Soil.

	Predicted Water	**Predicted Vegetation**	**Predicted Dry Soil**
Reference Water	**9**	**0**	**0**
Reference Vegetation	**0**	**1**	**3**
Reference Dry Soil	**1**	**0**	**14**

Discussion

The results obtained in this study demonstrate that the integration of Sentinel-1 C-band SAR imagery with Random Forest supervised classification, implemented within the Google Earth Engine cloud computing environment, constitutes a technically robust and operationally viable methodology for rapid flood extent mapping in complex metropolitan landscapes. The Overall Accuracy of 0.857 and Kappa coefficient of 0.742 achieved in this study are consistent with or exceed values reported in comparable SAR-based flood mapping studies in the scientific literature, supporting the validity of the proposed approach.

The superior performance of the water class relative to vegetation and dry soil reflects a fundamental characteristic of C-band SAR backscatter physics. Open water surfaces produce specular reflection of incident radar pulses, directing energy away from the sensor and generating consistently low backscatter values in both VV and VH polarizations. This physical mechanism creates a strong and stable spectral

contrast with rougher land surfaces, making water detection inherently well-suited to SAR-based classification regardless of the specific algorithm employed. The partial confusion observed between vegetation and dry soil classes is a recognized limitation of C-band SAR in environments where sparse or herbaceous vegetation produces insufficient volumetric scattering to be reliably distinguished from bare mineral surfaces. Future studies incorporating L-band SAR data, which penetrates vegetation canopy more deeply due to its longer wavelength, or multi-temporal SAR composites that exploit temporal backscatter changes associated with flood inundation, may further improve classification performance in these transitional land cover types.

The use of the RVImod index as an additional input feature alongside the VV and VH polarization bands contributed to enhanced inter-class separability by amplifying differential scattering responses across land cover types. The index formulation exploits the ratio between cross-polarized and co-polarized backscatter, which is sensitive to canopy volumetric scattering and surface moisture conditions. Its inclusion in the classifier input stack provides complementary information beyond individual polarization channels, consistent with findings in previous studies demonstrating the value of derived SAR indices for vegetation and flood mapping applications.

The cloud computing implementation of the full processing workflow in GEE represents a significant operational advantage relative to conventional desktop-based SAR processing pipelines. By eliminating the need for local data download, storage, and preprocessing, the GEE-based approach substantially reduces the time between satellite data acquisition and flood map generation, a critical factor in emergency response contexts where timely spatial information directly informs evacuation decisions, resource allocation, and search and rescue operations. The complete processing workflow, from image collection to classified output export, can be executed within minutes for regional-scale analysis, representing a meaningful advancement in operational flood monitoring capacity for disaster-prone regions such as southern Brazil.

It is important to acknowledge the limitations of the present study. The classification was performed using a single Sentinel-1 acquisition representing peak flood conditions, without incorporating pre-flood baseline imagery for change detection analysis. The inclusion of pre-event SAR data would enable the identification of newly inundated areas through backscatter change detection, potentially improving detection accuracy in partially flooded or transitional zones. Additionally, no independent field validation data were available to confirm the classification results at the ground level, and accuracy assessment was performed exclusively using the withheld validation subset of the original training samples. Future research incorporating field-validated inundation records or independent reference datasets such as flood extent polygons derived from aerial surveys or official emergency response reports would strengthen the reliability of accuracy estimates. Finally, the 206-point training sample, while sufficient to achieve strong classification performance in this study, may not fully capture the spectral variability of land cover

classes across the full extent of the metropolitan region, and expanding the training dataset in future iterations could further improve model robustness.

Conclusion

This study demonstrated the operational viability of combining Sentinel-1 C-band SAR imagery with Random Forest supervised classification, implemented within the Google Earth Engine cloud computing platform, for rapid flood extent mapping in the context of the May 2024 Rio Grande do Sul extreme flood event. The proposed methodology achieved an Overall Accuracy of 0.857 and a Cohen's Kappa coefficient of 0.742, indicating strong classification performance and reliable discrimination between open water, vegetation, and dry soil land cover classes across the Porto Alegre metropolitan region.

The results confirm that SAR-based flood mapping overcomes the fundamental observational constraint of optical remote sensing under cloud-covered conditions, which are inherently prevalent during active precipitation events. The all-weather acquisition capability of Sentinel-1, combined with the cloud computing scalability of Google Earth Engine and the adaptive classification performance of Random Forest, constitutes a robust and transferable framework for emergency flood mapping applicable to any flood-prone region with Sentinel-1 coverage encompassing virtually the entire global land surface.

The integration of the modified Radar Vegetation Index (RVImod) as a derived input feature enhanced inter-class spectral separability and contributed to overall classification accuracy, demonstrating the value of SAR-derived indices as complementary inputs to individual polarization bands in machine learning classification workflows. The complete processing pipeline, from data access to classified output export, can be executed within a cloud computing environment without requiring local data infrastructure, substantially reducing the time between satellite acquisition and actionable flood extent information, a critical advantage in disaster response contexts.

The methodology presented here is fully replicable and scalable, requiring only freely available Sentinel-1 data, open-access Google Earth Engine infrastructure, and a modest set of manually collected training samples. Its application is not limited to the specific event analyzed in this study but extends to any flood scenario captured within the Sentinel-1 temporal archive, supporting historical flood analysis, multi-event comparative studies, and near-real-time monitoring workflows. The open-source nature of the processing pipeline and the public availability of all input data support transparency, reproducibility, and community adoption in the remote sensing and disaster risk reduction communities.

Future research directions include the incorporation of pre-event SAR baseline imagery for change detection-based flood mapping, the exploration of multi-temporal SAR composites to improve classification in transitional inundation zones, the integration of digital elevation model data to constrain flood extent to topographically plausible areas, and the application of the methodology to other recent extreme flood

events in Brazil and internationally. The expansion of training datasets through active learning or transfer learning approaches may further improve model generalization across diverse flood environments and land cover contexts.

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Author

Carolina da Silva Ezequiel
Environmental Management Student | Remote Sensing & GIS Analyst
Unicesumar: Expected Graduation: November 2026
carolezeq15@gmail.com
GitHub: github.com/carolezeq-Analist
Portfolio: carolina-geo-folio.lovable.app/